Signed carriers, scalar recovery and a global P-192 degree constraint

Nonzero signed carrier F(X,Y) + Z H(X,Y)

Weights 1,1,3; global subgroup coverage or finite-chart total weight

Translate intersections -> elliptic subgroup

Explicit incidence inequality, adjunction and a free prime-order quotient

CM norm -> bounded public point equation

Projection degrees <=2d; [a]Q=[u]G and u^2=ab

Point hash -> scalar u/a, or degree exclusion

Verify the scalar point; retain compressed high-degree circuits as a separate interface

Construction, 2026-11-08, T-1928. This round constructs explicit signed bivariate carriers for small-scalar point actions, a conditional carrier-to-scalar decoder, and a global degree criterion linking such carriers to the CM norm. On actual P-192, affine and independently generated Jacobian point lists agree on all seven frozen target panels. The public target has no global signed carrier through weight2048 under the written criterion; the two labeled CM fixtures have the stronger bound2^44. These are scoped producer derivations and exact native checks, pending trusted review. [pcc:unverified]

Signed coordinates. Write X=x(P),Y=x(R),Z=y(P)y(R), with Z^2=f(X)f(Y), f(X)=X^3+aX+b. Use weights1,1,3 and reduce every polynomial to F(X,Y)+Z*H(X,Y). The relative sign retained inZ is the same information used by the Round58 pole extractor. A nonzero reduced pair is a nonzero rational function on E x E; universal multiples of the surface equation are excluded by this normal form. [pcc:unverified]

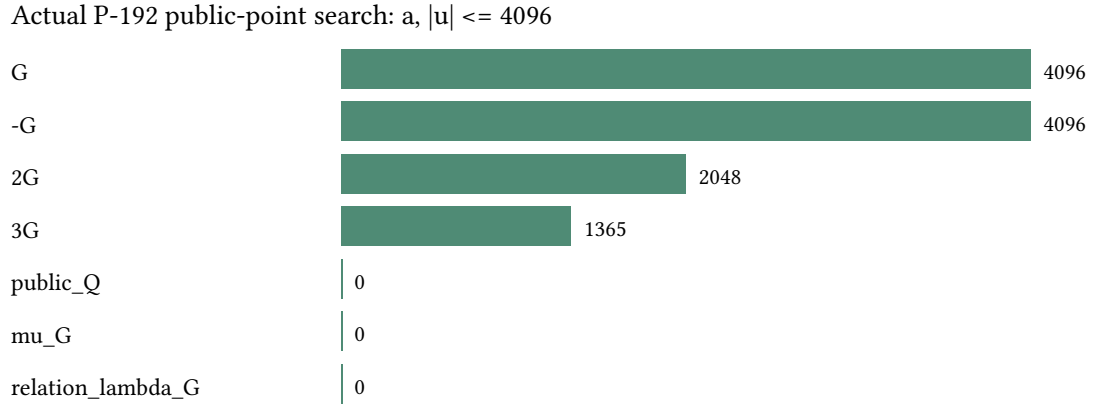
Constructive controls. For R=P and R=-P, use Y-X=0 together with Z-f(X)=0 or Z+f(X)=0, respectively. The signed equations have weight3; each two-equation system has total weight4. For R=2P, the tangent formula gives2Z-(3X^2+a)(X-Y)+2f(X)=0, of weight3. For R=3P, put N2=X^4-2aX^2-8bX+a^2 and use(Y-X)^2*(N2+4f(X)(Y+X))-4f(X)*(f(Y)+f(X)+2Z)=0, of weight6. The latter follows by imposing x(R-P)=x(2P). Both formulas also admit R=-P, so retain the guard Y!=X, checked through an inverse witness. On nonzero points of prime order r>5, the intended2P and3P branches satisfy that guard. These formulas retain an explicit point action rather than only a collision of curve identities. [pcc:unverified]

Global carrier theorem. Let Gamma=<(G,Q)> have prime order r!=p in E x E, and assume End(E_bar)=Z[pi], D=t^2-4p<0. Put s=ceil((r-1)/(4d)). If s(s-1)>8d^2(r-1),4d^2<r and16d^2<|D|, then a nonzero signed polynomial of weight<=d vanishing on every nonzero Gamma point forces1<=a<=2d,0<|u|<=2d and[a]Q=[u]G. More precisely u^2=ab for two elliptic projection degrees a,b<=2d. The full written proof and source dependencies are in carrier-theorem.md. [pcc:unverified]

Proof mechanism. The carrier has poles bounded by2d times the sum of the two origin fibers, giving a zero divisor of self-intersection8d^2 and at most4d components. One component contains at least s subgroup points. Counting their differences forces a nonzero subgroup translation to stabilize that component. Its normalization has a free degree-r etale quotient; adjunction and Riemann-Hurwitz force genus1 under4d^2<r. It is therefore an elliptic subgroup containing(G,Q). The projection-dual composite has norm ab, and4ab=(2u+tv)^2+|D|v^2. The final degree inequality forces v=0. The proof is a specialization of established translation/zero-estimate ideas, with explicit constants and signed-coordinate bookkeeping. [pcc:unverified]

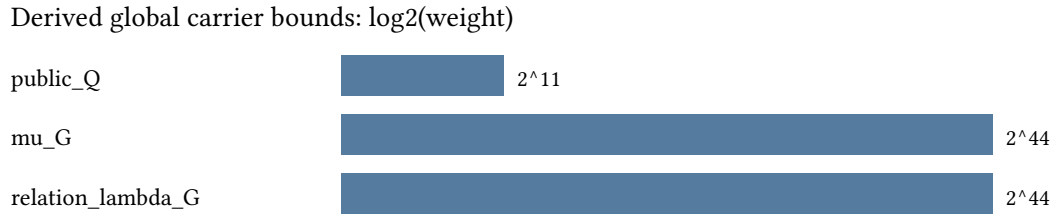
Finite charts and the decoder. If finitely many nonzero signed polynomial equations cover all nonzero Gamma points, their product remains nonzero and its weighted degree is at most the sum of their degrees; reduction by Z^2=f(X)f(Y) preserves that bound. Thus the theorem also constrains total finite-chart weight. Under the theorem's carrier promise, hash the points +/-[u]G for1<=u<=2d and scan[a]Q for1<=a<=2d. A match gives k=u/a modulo r; verify[k]G=Q. This requires O(d) point advances and O(d) storage, plus modular arithmetic and O(log r) group work for scalar verification. The carrier is a structural promise; its construction cost cannot be omitted. [pcc:unverified]

Actual P-192 bounded search. Keep all seven Round57 fixtures and search a,|u|<=4096. The full retained match counts are G:4096,-G:4096,2G:2048,3G:1365,public_Q:0,mu_G:0,relation_lambda_G:0. There are8192 signed table points and32768 point advances. Every point is checked on the curve. The alternate generator uses Jacobian mixed addition and batch normalization, rather than the producer's affine addition/inversion. Both complete outputs agree byte for byte. The decoder recovers1,r-1,2,3 on the four positive fixtures, verifies their points, and checks all11605 retained matches are scalar-consistent. The three zero-match rows remain outside this bounded-carrier promise. [pcc:unverified]



Counts of [a]Q=[u]G. Source: carriers.json; seven frozen fixtures; no runtime inference.

Larger labeled-fixture bounds. For d=2^44 set B=2d=2^45. The native program verifies all three geometric inequalities exactly. It also verifies mu!=1 and mu^5=1 modulo r. A bounded relation u=a*mu would make r divide the positive integer Phi5(u,a)=u^4+u^3*a+u^2*a^2+u*a^3+a^4, whose absolute value is at most5B^4<r. This is impossible. For the horizontal-relation label lambda, let L=r-lambda=261576852865688056623483890096684927878131 7. Exact inequalities L>B and B*L+B<r place every a*lambda outside the signed interval of widthB. Hence neither labeled fixture has a global signed carrier or finite-chart collection of total weight<=2^44. No large-box point enumeration is used. [pcc:unverified]



Written geometric criterion + exact scalar certificates; global equations and total finite-chart weight.

What the bounds select next. The small-degree signed controls provide explicit global action descriptions; the two existing CM labels cannot supply a low-weight description of that kind. The standard P-192 trace and saved Round4 fundamental-discriminant certificate supply the endomorphism-ring hypothesis. This is not an arithmetic-circuit lower bound: high-degree expressions such as X^p-X or the Frobenius normalizer can be cheap to evaluate. Nor does the theorem exclude an equation valid only on a baby subset. The next construction should therefore target a compressed high-degree norm or a subset representation, while retaining the signed mark and accounting for the cost of obtaining it from public G,Q. Universal rational-point equations alone carry no target-specific mark. [pcc:unverified]

Exact checks and custody. Four positive carrier panels cover192 P-192 sample pairs, with192 wrong-relative-sign rejections,96 guard witnesses and96 checks of the excluded minus-P branch. The frozen point search, Jacobian replay and fresh-copy outputs agree; wrong-sign and corrupted-match-count mutations reject with exit101. The subsequent decoder is covered by its before-data amendment and fresh replay. Source/scope commits precede their affected checks. validation.json records the required broad library check and actual failures. No timing or statistical claim, full P-192 trace, cloud job or generator launch is included. The written geometric proof is not a Lean formalization. [pcc:unverified]

Sources checked (KB unavailable). MCP tools are not loaded and the migrated repository wrapper is absent. Brown1992 section2 Theorem2.1, https://www.numdam.org/item/10.24033/asens.1646.pdf , places the method within arbitrary-characteristic zero estimates and credits Masser-Wustholz. Milne Corollary2.2/Remark12.13, https://www.jmilne.org/math/xnotes/AVs.pdf , give rigidity and translation invariance of divisor classes. Vakil Exercise20.2.B(a), printed page581, https://math.stanford.edu/~vakil/216blog/FOAGjul 2724public.pdf , supplies arithmetic-genus adjunction. Stacks0C1B, https://stacks.math.columbia.edu/tag/0C1B , supplies Riemann-Hurwitz. These support the written specialization; no novelty claim for the standard zero-estimate mechanism is made. [pcc:unverified]